WORKSHEET - NAPLAN Year 9

Number and Algebra

Patterns (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes *(based on NAPLAN allocation of ~1.25 min/mark)*

Questions

1. Number and Algebra, NAP-20263

Ethan writes the sequence:

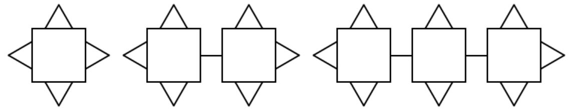
$7, 0.7, 0.07, 0.007, \ldots$

Describe the rule he uses to write the next number in this sequence.

- add 6.3
- subtract 6.3
- multiply by 0.01
- divide by 0.1
- multiply by 10
- divide by 10

2. Number and Algebra, NAP-90993

Tim produced a pattern with large squares and small triangles.



Shape no.	1	2	3
Large squares	1	2	3
Small triangles	4	6	8

If Tim continues this pattern how many small triangles will be in the 12th shape?

6 12 26 30

3. Number and Algebra, NAP-32122

Charlie swam 250 lengths of a 50 metre pool in 5 days.

After the first day, he swam 5 more laps each day than the day before.

How many laps did he swim on the first day?

35 40 45 50

4. Number and Algebra, NAP-26210

Barry drives a barge that transports sand.

The table below shows the volume of sand, measured in cubic metres, of different sand masses, measured in tonnes.

Sand Volume	Sand Mass
20	30
50	75
100	150
200	300

Barry's barge has 84 cubic metres of sand loaded on it.

How many tonnes of sand are on his barge?

56 tonnes   58 tonnes   126 tonnes   136 tonnes

5. Number and Algebra, NAP-08632

Stewart uses the formulas below to estimate the population of koalas in a habitat over four years.

$$\text{Population (Year 1)} = 5000$$

$$\text{Population (Year 2)} = \text{Year 1} + \frac{\text{Year 1}}{10}$$

$$\text{Population (Year 3)} = \text{Year 2} + \frac{\text{Year 2}}{10}$$

$$\text{Population (Year 4)} = \text{Year 3} + \frac{\text{Year 3}}{10}$$

What is the estimated population of koalas in year 4?

6. Number and Algebra, NAP-38068

Richard is ascending in a hot air balloon.

The temperature drops by the same amount every 50 metres that Richard ascends.

At ground level the temperature is 31 °C.

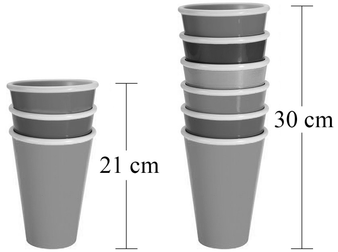
At a height of 600 metres, the temperature is –5 °C.

What is the air temperature at a height of 200 metres?

16 °C   18 °C   19 °C   22 °C

7. Number and Algebra, NAP-49763

Angela stacks up identical cups.



A stack of 3 cups is 21 cm high.

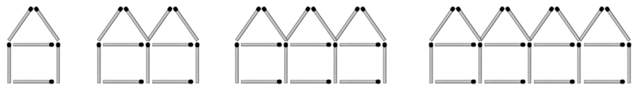
A stack of 6 cups is 30 cm high.

What height will a stack of 10 similar cups be?

70 cm   54 cm   50 cm   42 cm

8. Number and Algebra, NAP-02717

Mike uses match sticks to make a pattern of houses.



Number of houses	1	2	3	4
Number of sticks	6	11	16	21

When a shape consists of 7 houses, how many sticks will the shape have?

Worked Solutions

1. Number and Algebra, NAP-20263

$$7 \div 10 = 0.7$$
$$0.7 \div 10 = 0.07$$
$$0.07 \div 10 = 0.007$$

$\therefore$ Rule: divide by 10.

2. Number and Algebra, NAP-90993

Let s = Number of large squares

$\therefore$ Number of triangles in the 12th shape

$$= (s \times 2) + 2$$
$$= 12 \times 2 + 2$$
$$= 24 + 2$$
$$= 26$$

3. Number and Algebra, NAP-32122

Average laps per day

$$= \frac{250}{5} = 50$$

$\Rightarrow$ Laps swam each of the 5 days:

40, 45, 50, 55, 60

$\therefore$ He swam 40 laps on the 1st day.

4. Number and Algebra, NAP-26210

Tonnes in 1 cubic metre of sand

$$= \frac{30}{20} = 1.5$$

$\therefore$ Tonnes in 84 cubic metres

$$= 84 \times 1.5$$
$$= 126 \text{ tonnes}$$

5. Number and Algebra, NAP-08632

$$\text{Year } 2 = 5000 + \frac{5000}{10} = 5500$$

$$\text{Year } 3 = 5500 + \frac{5500}{10} = 6050$$

$$\text{Year } 4 = 6050 + \frac{6050}{10} = 6655$$

6. Number and Algebra, NAP-38068

There are 12×50 m distances in 600 m.

Temperature drop each 50 m

$$= \frac{31 - (-5)}{12}$$

$$= \frac{36}{12}$$

$$= 3$$

$\therefore$ Temperature at 200 m

$$= 31 - (4 \times 3)$$

$$= 19\ ^\circ\text{C}$$

7. Number and Algebra, NAP-49763

Height difference between 3 cups and 6 cups

$$= 30 - 21$$

$$= 9 \text{ cm}$$

$\Rightarrow$ Each extra cup adds 3 cm.

Since 6 cups are 30 cm,

$\therefore$ Height of 10 cups (an extra 4)

$$= 30 + (4 \times 3)$$

$$= 42 \text{ cm}$$

8. Number and Algebra, NAP-02717

1st house - 6 sticks

Add 5 sticks for each extra houses

5 houses - 26

6 houses - 31

7 houses - 36